

A Numerical Example of Attention Mechanism in a Transformer

Ping Wang

April 2026

[Tutorials](#) • [Datasets](#) • [Exhibits](#) • [Colab Notebooks](#)

1 Four token IDs

$$n_{\text{embd}} = d_{\text{model}} = 4, \quad n_{\text{head}} = 2, \quad d_k = d_v = 2$$

Assume four token IDs:

$$[2, 0, 3, 1]$$

Using the column-vector convention, let the learned token embedding matrix be

$$E = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} x_1 = E[:, 2] &= \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, & x_2 = E[:, 0] &= \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \\ x_3 = E[:, 3] &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, & x_4 = E[:, 1] &= \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}. \end{aligned}$$

Thus

$$X = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

2 Positonal embeddins

For sinusoidal positional encoding,

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right),$$

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right).$$

Since

$$d_{\text{model}} = 4,$$

we have

$$p_{pos} = \begin{bmatrix} \sin(pos) \\ \cos(pos) \\ \sin(pos/100) \\ \cos(pos/100) \end{bmatrix}.$$

For positions 0, 1, 2, 3,

$$p_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix},$$

$$p_1 \approx \begin{bmatrix} 0.8415 \\ 0.5403 \\ 0.0100 \\ 1.0000 \end{bmatrix},$$

$$p_2 \approx \begin{bmatrix} 0.9093 \\ -0.4161 \\ 0.0200 \\ 0.9998 \end{bmatrix},$$

$$p_3 \approx \begin{bmatrix} 0.1411 \\ -0.9900 \\ 0.0300 \\ 0.9996 \end{bmatrix}.$$

The input representation for token i is

$$h_i = x_i + p_i.$$

Therefore,

$$h_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix},$$

$$h_2 \approx \begin{bmatrix} 1.8415 \\ 0.5403 \\ 1.0100 \\ 1.0000 \end{bmatrix},$$

$$h_3 \approx \begin{bmatrix} 0.9093 \\ -0.4161 \\ 1.0200 \\ 1.9998 \end{bmatrix},$$

$$h_4 \approx \begin{bmatrix} 0.1411 \\ 0.0100 \\ 0.0300 \\ 1.9996 \end{bmatrix}.$$

Stacking the representations as columns,

$$H = \begin{bmatrix} 1 & 1.8415 & 0.9093 & 0.1411 \\ 2 & 0.5403 & -0.4161 & 0.0100 \\ 0 & 1.0100 & 1.0200 & 0.0300 \\ 1 & 1.0000 & 1.9998 & 1.9996 \end{bmatrix}.$$

3 Head 1: Compute Q, K, V

Since

$$n_{\text{head}} = 2,$$

each head has

$$d_k = d_v = \frac{n_{\text{embd}}}{n_{\text{head}}} = 2.$$

For head 1, choose

$$W_Q^{(1)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix},$$

$$W_K^{(1)} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$W_V^{(1)} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Then

$$Q^{(1)} = W_Q^{(1)}H, \quad K^{(1)} = W_K^{(1)}H, \quad V^{(1)} = W_V^{(1)}H.$$

Hence

$$Q^{(1)} = \begin{bmatrix} 1 & 1.8415 & 0.9093 & 0.1411 \\ 2 & 0.5403 & -0.4161 & 0.0100 \end{bmatrix},$$

$$K^{(1)} = \begin{bmatrix} 0 & 1.0100 & 1.0200 & 0.0300 \\ 1 & 1.0000 & 1.9998 & 1.9996 \end{bmatrix},$$

$$V^{(1)} = \begin{bmatrix} 1 & 2.8515 & 1.9293 & 0.1711 \\ 3 & 1.5403 & 1.5837 & 2.0096 \end{bmatrix}.$$

4 Attention scores

For token 1,

$$q_1^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad k_1^{(1)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad v_1^{(1)} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

The raw attention-score matrix is

$$R^{(1)} = (Q^{(1)})^T K^{(1)}.$$

The scaled score matrix is

$$S^{(1)} = \frac{(Q^{(1)})^T K^{(1)}}{\sqrt{d_k}} = \frac{(Q^{(1)})^T K^{(1)}}{\sqrt{2}}.$$

Approximately,

$$S^{(1)} \approx \begin{bmatrix} 1.4142 & 2.1283 & 3.5494 & 2.8490 \\ 0.3821 & 1.6972 & 2.0922 & 0.8030 \\ -0.2943 & 0.3552 & 0.0674 & -0.5691 \\ 0.0071 & 0.1079 & 0.1159 & 0.0171 \end{bmatrix}.$$

5 Softmax and weighted sum of V

The attention weights are

$$A^{(1)} = \text{softmax}(S^{(1)}),$$

where softmax is applied row-wise.

Approximately,

$$A^{(1)} \approx \begin{bmatrix} 0.0637 & 0.1301 & 0.5388 & 0.2674 \\ 0.0849 & 0.3163 & 0.4695 & 0.1293 \\ 0.1957 & 0.3747 & 0.2810 & 0.1487 \\ 0.2363 & 0.2614 & 0.2635 & 0.2387 \end{bmatrix}.$$

For token 1,

$$z_1^{(1)} = \sum_{j=1}^4 A_{1j}^{(1)} v_j^{(1)}.$$

Thus,

$$z_1^{(1)} \approx \begin{bmatrix} 1.5199 \\ 1.7821 \end{bmatrix}.$$

For all tokens,

$$Z^{(1)} = V^{(1)}(A^{(1)})^T.$$

Approximately,

$$Z^{(1)} \approx \begin{bmatrix} 1.5199 & 1.9147 & 1.8315 & 1.5310 \\ 1.7821 & 1.7453 & 1.9079 & 2.0087 \end{bmatrix}.$$

6 Head 2

For head 2, choose a different set of learned matrices:

$$W_Q^{(2)} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix},$$

$$W_K^{(2)} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix},$$

$$W_V^{(2)} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Then

$$Q^{(2)} = W_Q^{(2)} H, \quad K^{(2)} = W_K^{(2)} H, \quad V^{(2)} = W_V^{(2)} H.$$

The scaled attention scores are

$$S^{(2)} = \frac{(Q^{(2)})^T K^{(2)}}{\sqrt{2}},$$

and the attention weights are

$$A^{(2)} = \text{softmax}(S^{(2)}).$$

The head-2 output is

$$Z^{(2)} = V^{(2)} (A^{(2)})^T.$$

Suppose numerically,

$$Z^{(2)} \approx \begin{bmatrix} 0.4444 & 0.8139 & 0.7084 & 0.4158 \\ 1.0637 & 1.2597 & 1.2281 & 1.1201 \end{bmatrix}.$$

7 Multi-head concatenation

For token 1,

$$z_1^{(1)} = \begin{bmatrix} 1.5199 \\ 1.7821 \end{bmatrix}, \quad z_1^{(2)} = \begin{bmatrix} 0.4444 \\ 1.0637 \end{bmatrix}.$$

The multi-head output before the final output projection is the concatenation

$$z_1 = \text{Concat} \left(z_1^{(1)}, z_1^{(2)} \right),$$

so

$$z_1 = \begin{bmatrix} 1.5199 \\ 1.7821 \\ 0.4444 \\ 1.0637 \end{bmatrix} \in \mathbb{R}^{4 \times 1}.$$

For all tokens,

$$Z_{\text{concat}} = \text{Concat} \left(Z^{(1)}, Z^{(2)} \right)$$

and

$$Z_{\text{concat}} \approx \begin{bmatrix} 1.5199 & 1.9147 & 1.8315 & 1.5310 \\ 1.7821 & 1.7453 & 1.9079 & 2.0087 \\ 0.4444 & 0.8139 & 0.7084 & 0.4158 \\ 1.0637 & 1.2597 & 1.2281 & 1.1201 \end{bmatrix}.$$